If $x = f(t)$, $y = g(t)$, then $f^{-1}(x) = t$, so

$y = g(f^{-1}(x))$. Thus $\dfrac{dy}{dx} = g'(f^{-1}(x)) \cdot (f^{-1})'(x)$.

Now $(f^{-1})'(x) = \dfrac{1}{f'(f^{-1}(x))}$, so

$$\frac{dy}{dx} = \frac{g'(f^{-1}(x))}{f'(f^{-1}(x))}.$$ Since $t = f^{-1}(x)$,

$$\frac{dy}{dx} = \frac{g'(t)}{f'(t)}.$$

<u>Example</u> $x = t^3$, $y = \sin t$. Then

$t = x^{1/3}$, so $y = \sin(x^{1/3})$. Hence

$$\frac{dy}{dx} = \cos(x^{1/3}) \cdot \frac{1}{3} x^{-2/3} = \frac{\cos(x^{1/3})}{3(x^{1/3})^2}$$

$$= \frac{\cos t}{3t^2}.$$

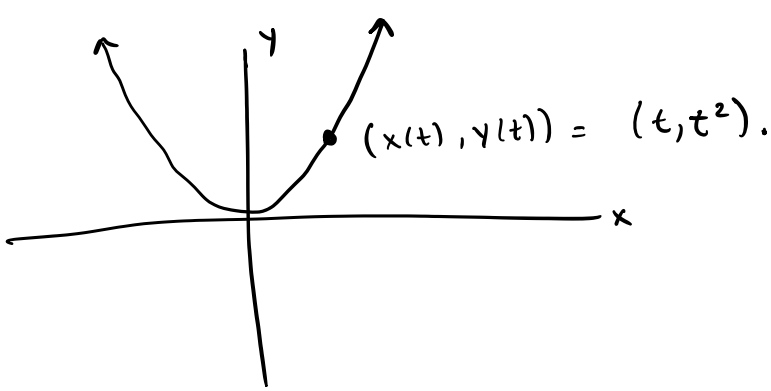
In other words, if x and y are functions of a common parameter t, and if you want to know

how y changes when x changes $\left(\frac{dy}{dx}\right)$, just compute $\frac{dy}{dt}$, $\frac{dx}{dt}$, and then

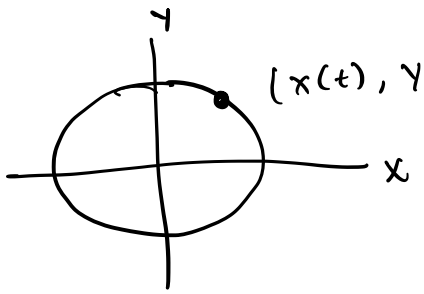
$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}.$$

Now when x, y are functions of t, the set of all points $(x(t), y(t))$ is a curve in the plane.

<u>Example</u> $x(t) = t$, $y(t) = t^2$. As t ranges over all real numbers, the points $(x(t), y(t))$ trace out a parabola



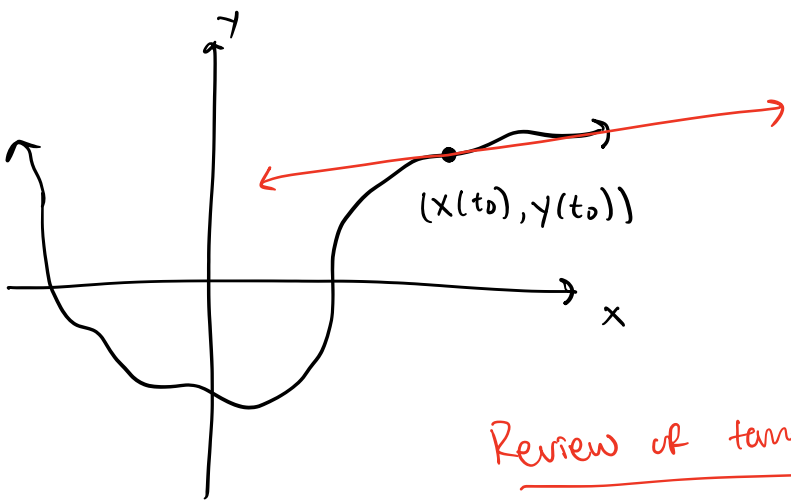
<u>Example</u> $x(t) = \cos t$, $y(t) = \sin t$. This traces out a circle as t goes from 0 to 2π



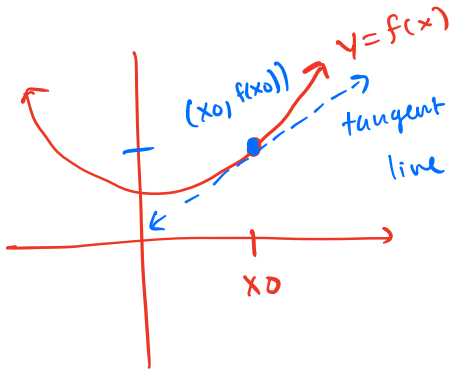
$(x(t), y(t)) = (\cos t, \sin t)$.

Let $x(t)$, $y(t)$ be given functions. Fix t_0.

What is the tangent line to the curve at t_0?



<u>Review of tangent lines</u>



Given $y = f(x)$, the tangent line to the curve defined by at $x = x_0$ is given by the equation

$$y = f(x_0) + f'(x_0)(x - x_0)$$

The above fact about tangent lines works when we can write y as a function of x.

Thus, given $x = f(t)$, $y = g(t)$, if we can invert f to get $t = f^{-1}(x)$, then $y = g(f^{-1}(x))$ ie, y can be written as a function of x. The tangent line to our curve at a particular point $(x_0, y_0) = (f(t_0), g(t_0)) = (x_0, g(f^{-1}(x_0)))$ is then given by the equation

$$y = g(f^{-1}(x_0)) + (g \circ f^{-1})'(x_0)(x - x_0).$$

Using the fact that we showed $(g \circ f^{-1})'(x_0) = \dfrac{g'(t_0)}{f'(t_0)}$ at the beginning, we have that the tangent line to the curve defined by $x = f(t)$, $y = g(t)$ at $t = t_0$ is given by

$$\boxed{y = g(t_0) + \frac{g'(t_0)}{f'(t_0)}(x - f(t_0)).}$$

<u>Example</u> $x = \underbrace{2t}_{f(t)}, \quad y = \underbrace{\sin(t)}_{g(t)}.$

Find the tangent line to the curve above

at $t = 1$.

"Harder" Way: $x = 2t \rightarrow t = x/2 \rightarrow y = \sin(x/2)$

$\rightarrow$ tangent line at $(x, y) = (2, \sin 1)$ is

$$y = \sin 1 + \underbrace{\frac{1}{2}\cos(1)}_{\frac{dy}{dx}(2)}(x - 2)$$

"Easier Way": Tangent line is

$$y = \underbrace{\sin 1}_{g(1)} + \underbrace{\frac{1}{2}\cos(1)}_{\frac{g'(1)}{f'(1)}}(x - \underbrace{2}_{f(1)})$$

The last example didn't really show how useful our method is. Let's look at a more

complicated case:

<u>Example</u> $x = \underbrace{\sin(\ln(t))}_{f(t)}, \quad y = \underbrace{e^{t^2}}_{g(t)}$

Find tangent line at $t = 1$.

Easy way: $f(1) = 0, \quad g(1) = e,$

$f'(t) = \dfrac{\cos(\ln(t))}{t} \rightarrow f'(1) = 1$

$g'(t) = 2te^{t^2} \rightarrow g'(1) = 2e$

$\therefore \quad y = e + 2ex$ is the tangent line to the curve at $t = 1$.

Hard way: $x = \sin(\ln(t)) \rightarrow$

$e^{\sin^{-1}(x)} = t \rightarrow y = \underbrace{e^{(e^{\sin^{-1}(x)})^2}}_{h(x)}$

$t = 1 \rightarrow x = 0, \; y = e.$ The tangent line to the curve when $t = 1$ is given by

$y = e + h'(0)x$. Let's compute $h'(x)$

to emphasize how painful this is:

$$h'(x) = \left(\left(e^{\sin^{-1}(x)}\right)^2\right)' e^{\left(e^{\sin^{-1}(x)}\right)^2}$$

$$= e^{\left(e^{\sin^{-1}(x)}\right)^2} \cdot \left[2e^{\sin^{-1}(x)} \cdot \left(\sin^{-1}(x)\right)'\right]$$

what is $\left(\sin^{-1}(x)\right)'$? : $x = \sin(\sin^{-1}(x)) \rightarrow$

$$1 = \cos(\sin^{-1}(x)) \cdot \left(\sin^{-1}(x)\right)'$$

$$\rightarrow \left(\sin^{-1}(x)\right)' = \frac{1}{\cos(\sin^{-1}(x))} \rightarrow$$

$$h'(x) = e^{\left(e^{\sin^{-1}(x)}\right)^2} \cdot 2e^{\sin^{-1}(x)} \cdot \frac{1}{\cos(\sin^{-1}(x))} \rightarrow$$

$$h'(0) = e^{(e^0)^2} \cdot 2e^0 \cdot \frac{1}{\cos(0)} = 2e,$$ which

gives us the same result as the first way does.

Hopefully this shows why way # 1 is more useful.

<u>Summary</u> $x = f(t),\ y = g(t) \longrightarrow$

1. $y = g(f^{-1}(x))$

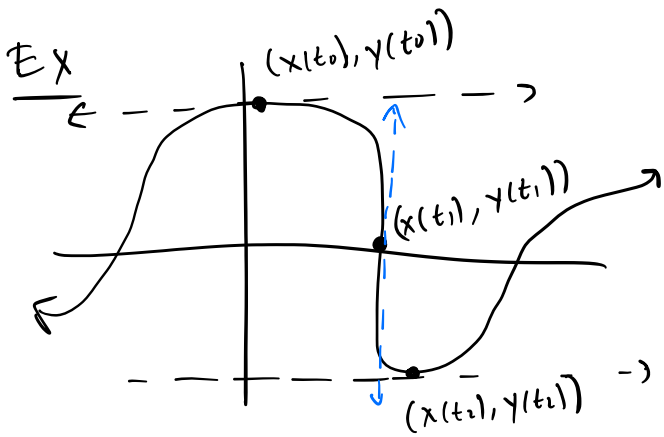
2. $\dfrac{dy}{dx}(x) = \dfrac{dy}{dx}(f(t)) = \dfrac{g'(t)}{f'(t)}$

3. Tangent line to curve $x = f(t),\ y = g(t)$ at $t = t_0$ is

$$y = g(t_0) + \frac{g'(t_0)}{f'(t_0)}\left(x - f(t_0)\right)$$

Now given a curve $x = f(t),\ y = g(t)$

Ex



at $t = t_0$ and $t = t_2$ the tangent to the curve is horizontal which means its slope is zero!

Since the slope of the tangent to the curve is given by

$\dfrac{g'(t)}{f'(t)}$, we conclude that $g'(t_0) = 0$ and

$g'(t_2) = 0$.

ie $\boxed{\text{Horizontal tangent line at } t = t_0 \iff g'(t_0) = 0}$

Similarly, the tangent line at $t = t_1$ is vertical, so the slope $\dfrac{g'(t_1)}{f'(t_1)}$ is undefined.

This means that $f'(t_1) = 0$. ie

$\boxed{\text{Vertical tangent line at } t = t_0 \iff f'(t_0) = 0}$

<u>Summary</u> $x = f(t),\ y = g(t) \longrightarrow$

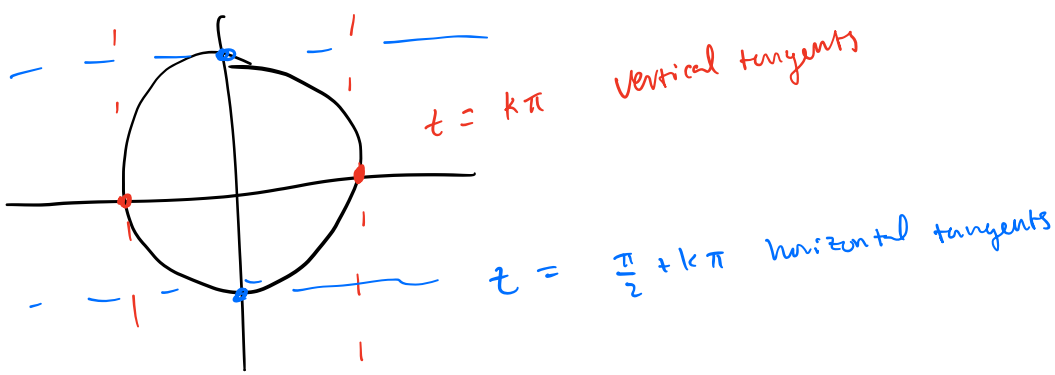
Vertical tangent at $t = t_0$ $\iff f'(t_0) = 0$

Horizontal tangent at $t = t_0$ $\iff g'(t_0) = 0$

Example $\quad x = \underbrace{\cos(t)}_{f(t)}, \quad y = \underbrace{\sin t}_{g(t)}$

Vertical tangents : $f'(t) = -\sin t = 0$

$\longrightarrow \quad \sin t = 0 \quad \longrightarrow \quad t = k\pi$, k an integer

Horizontal tangents : $g'(t) = \cos t = 0 \quad \longrightarrow$

$t = \frac{\pi}{2} + k\pi$, k integer